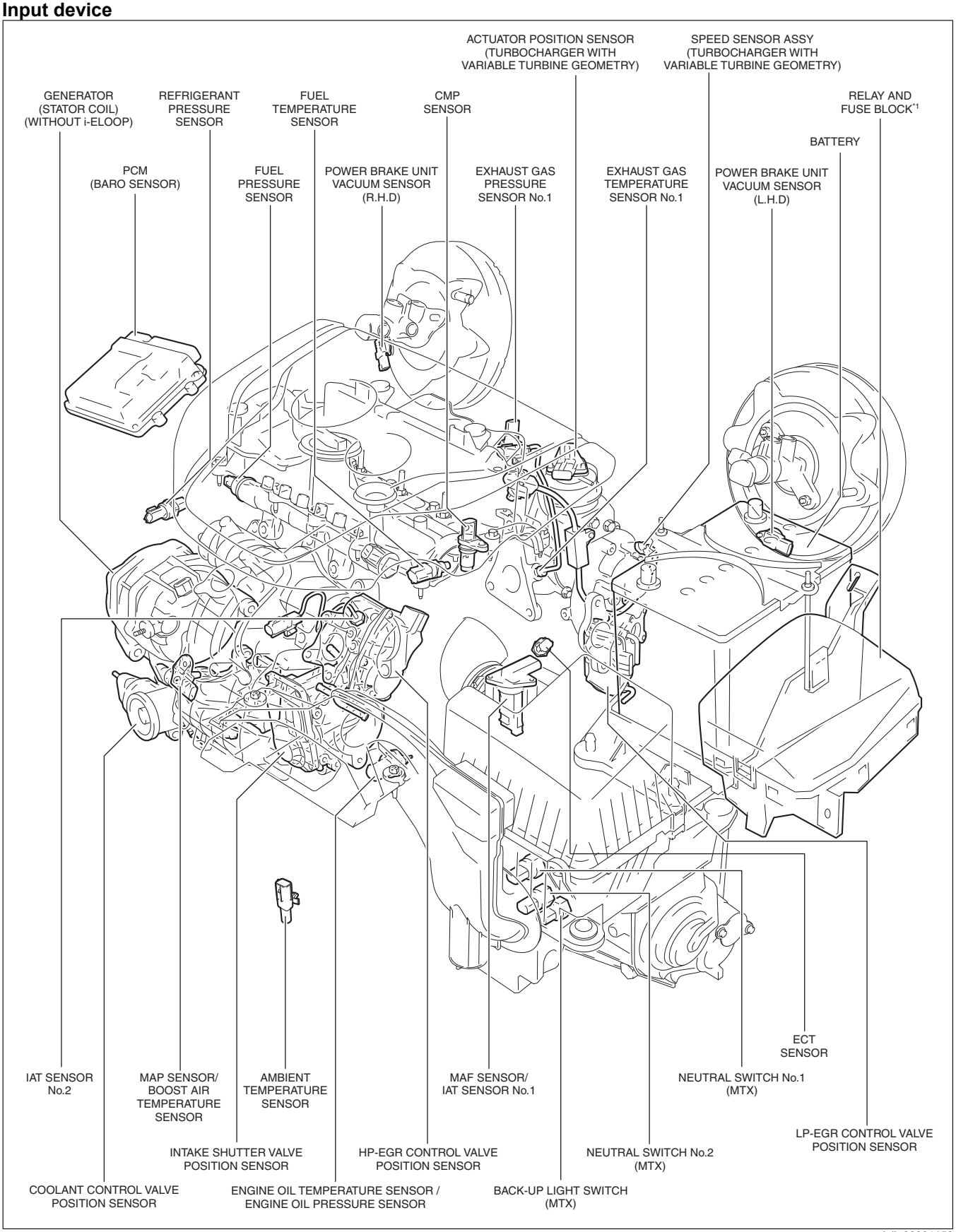
ENGINE CONTROL SYSTEM [SKYACTIV-D 1.5]

id0140z5139900

Outline

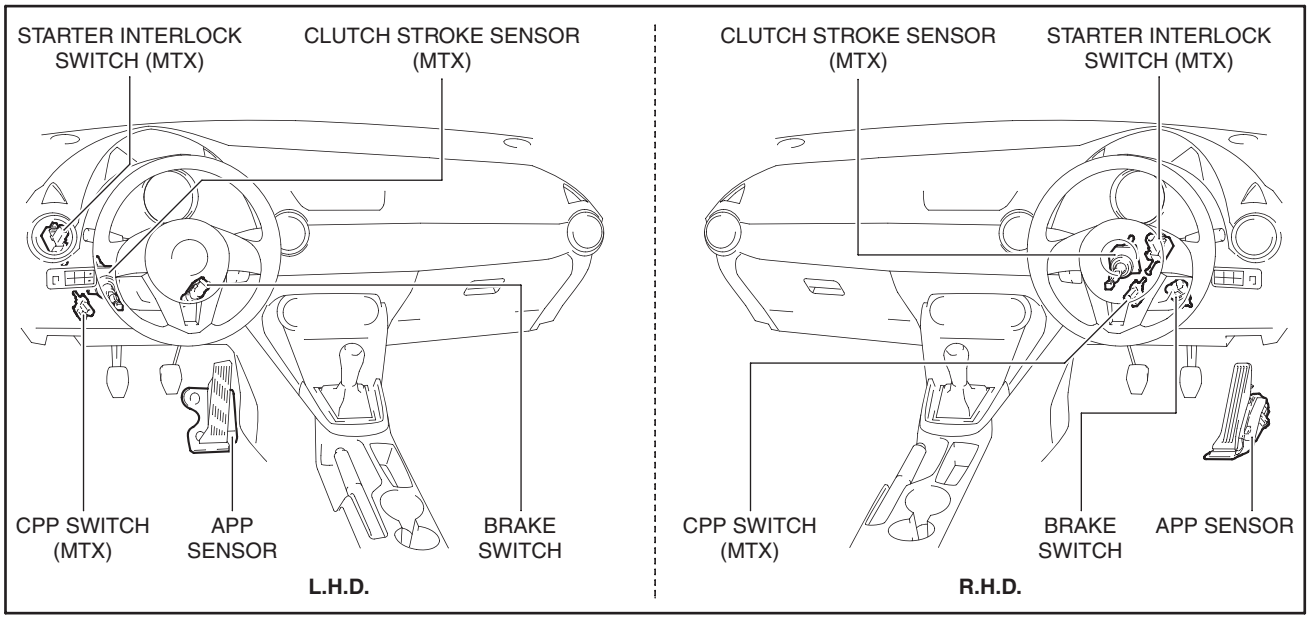
- Glow control has been adopted to improve engine startability and diesel particulate filter regeneration performance.
  - — Glow control module adopted
- Engine hydraulic pressure switching control has been adopted reducing engine load.
  - — Engine oil solenoid valve adopted
- Coolant control valve control has been adopted for improved fuel economy.
  - — Coolant control valve adopted
- Exhaust gas recirculation (EGR) system has been adopted which achieves cleaner exhaust emissions and improved fuel efficiency.
- Boost control has been adopted for improved vehicle acceleration performance, fuel economy, and environmental performance.
  - — Variable geometry turbocharger adopted
  - — Speed sensor assy (turbocharger with variable turbine geometry) adopted
- Generator output control adopted, fuel economy/idling stability improved.
  - — Current sensor adopted
- LIN communication has been adopted to the current sensor for simplified wiring harnesses.
- i-stop control has been adopted for improved fuel economy and reduce exhaust gas and idling noise.
- DC-DC converter control has been adopted for improved power supply stability. (without i-ELOOP)
  - — DC-DC converter adopted
- i-ELOOP has been adopted for improved electrical power generation during deceleration. (with i-ELOOP) (See i-ELOOP [i-ELOOP].)
  - — DC-DC converter (i-ELOOP) adopted

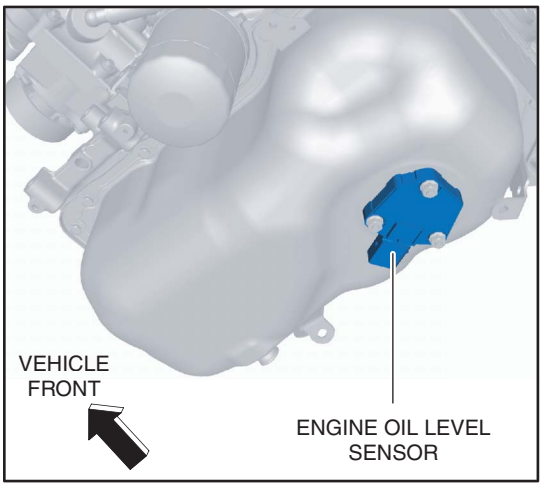
Structural View
Input device

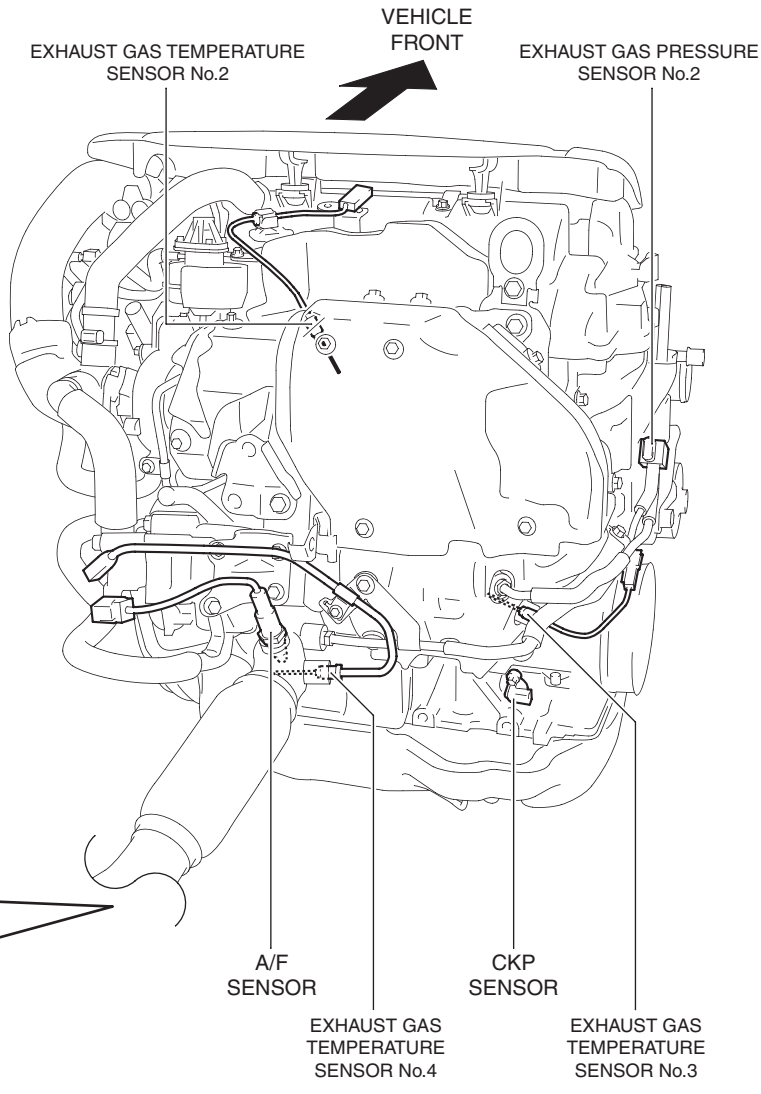


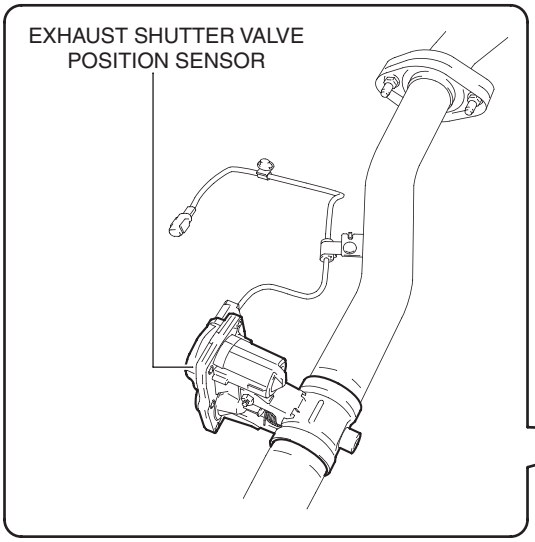
adejjn00004153

*1 : Check connector, IG1 relay



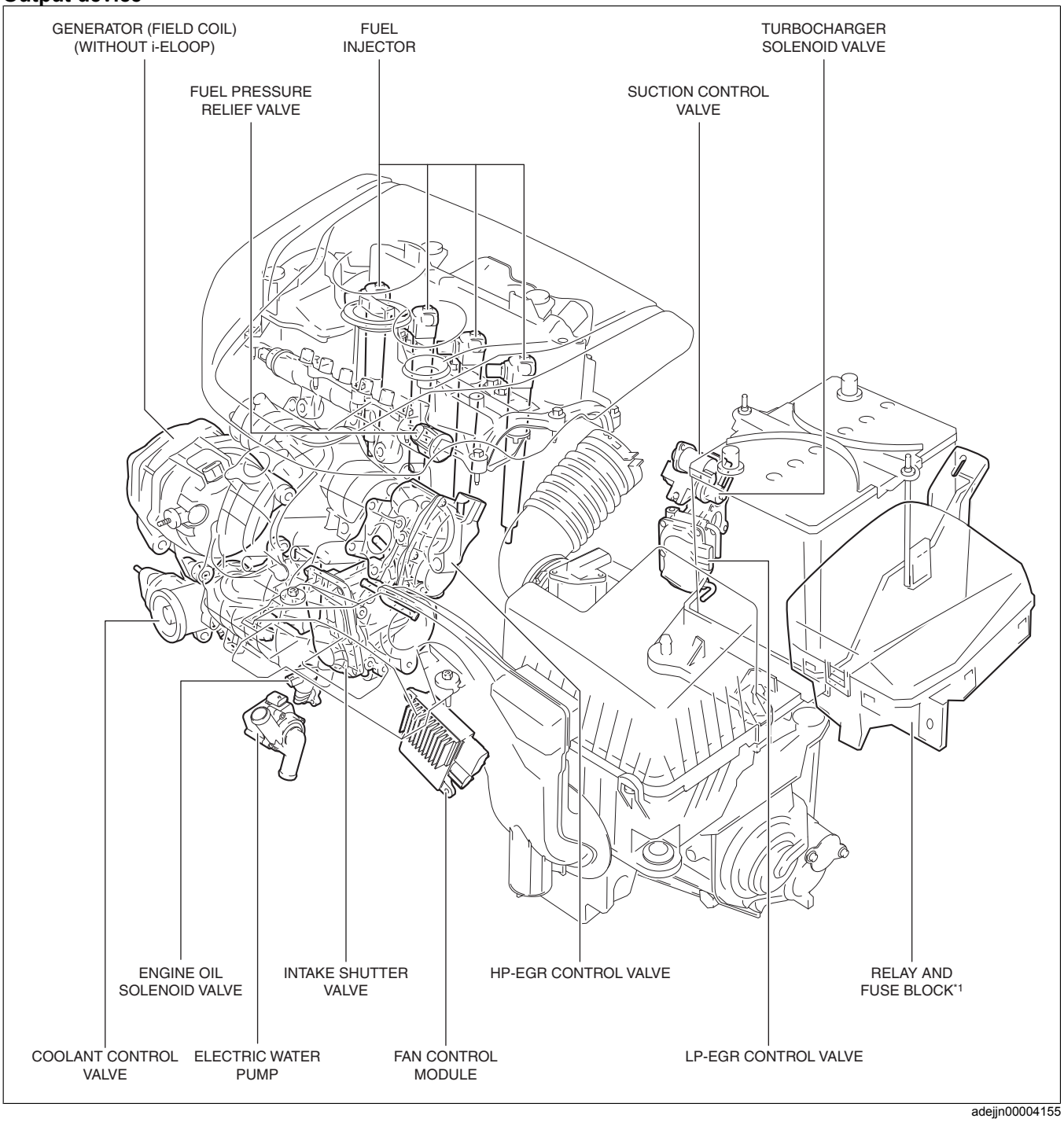






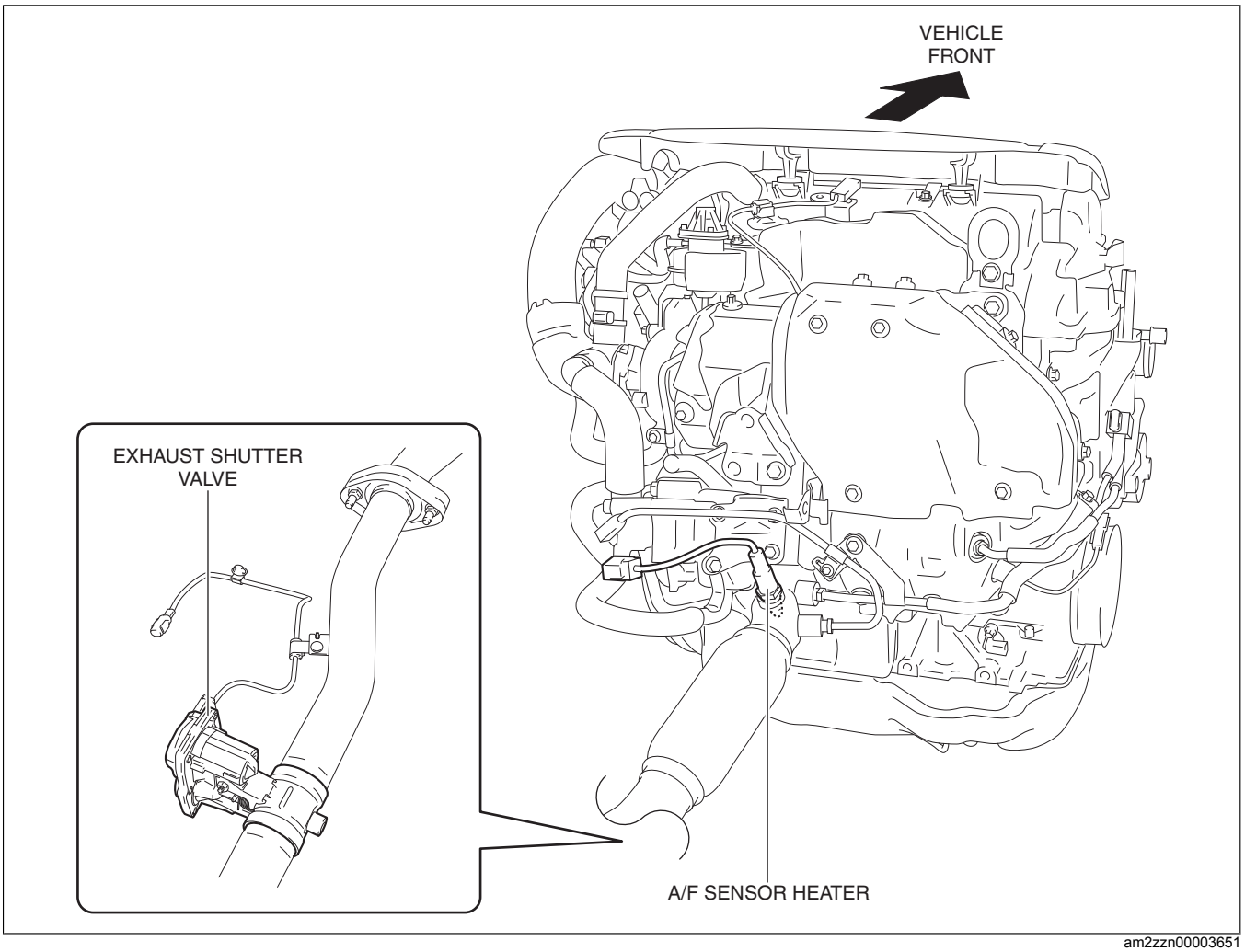
am2zzn00003650

Output device



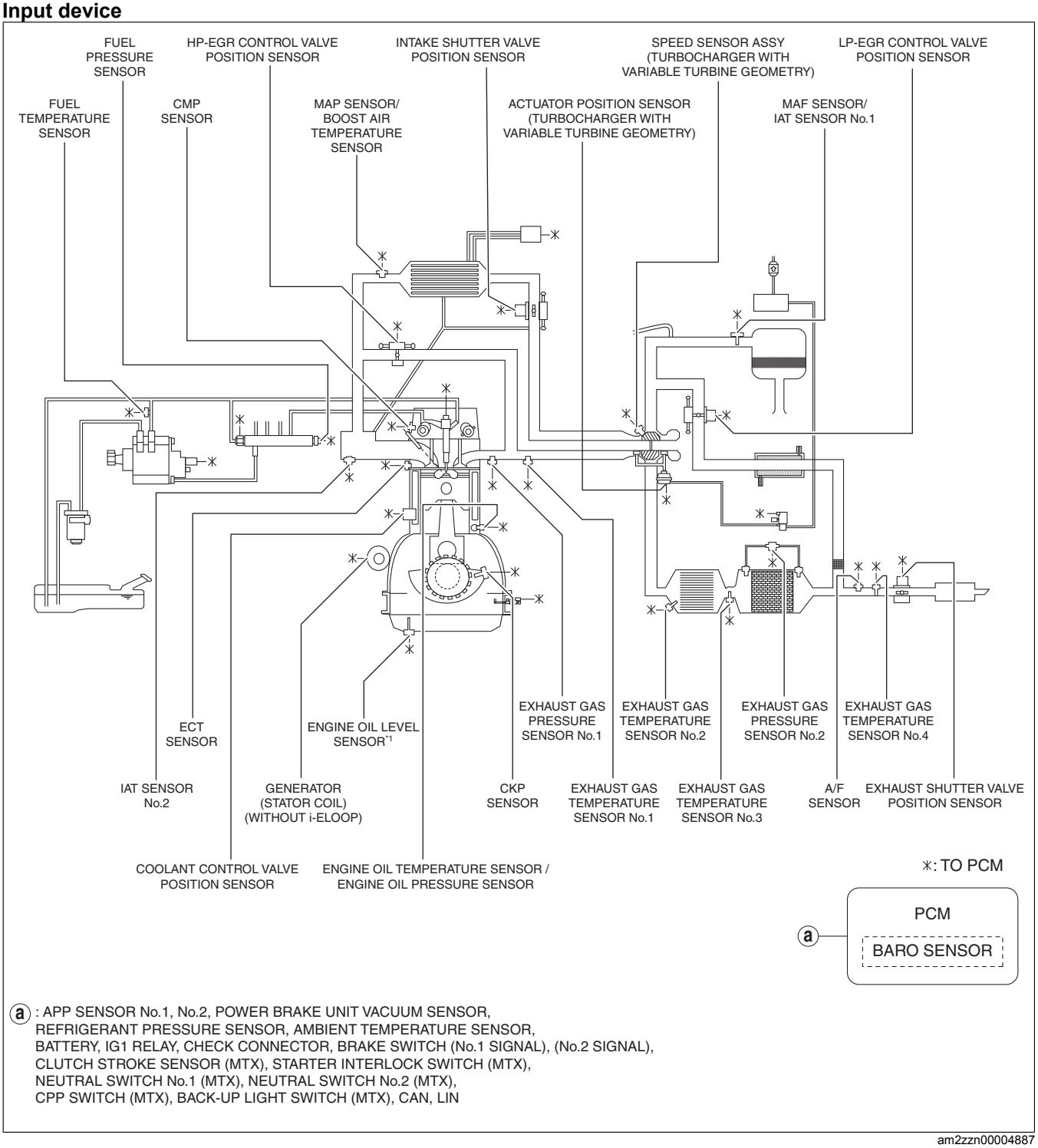
adejjn00004155

*1 : Main relay, cooling fan relay, A/C relay, starter relay, blow-by heater relay



am2zzn00003651

System Diagram
Input device

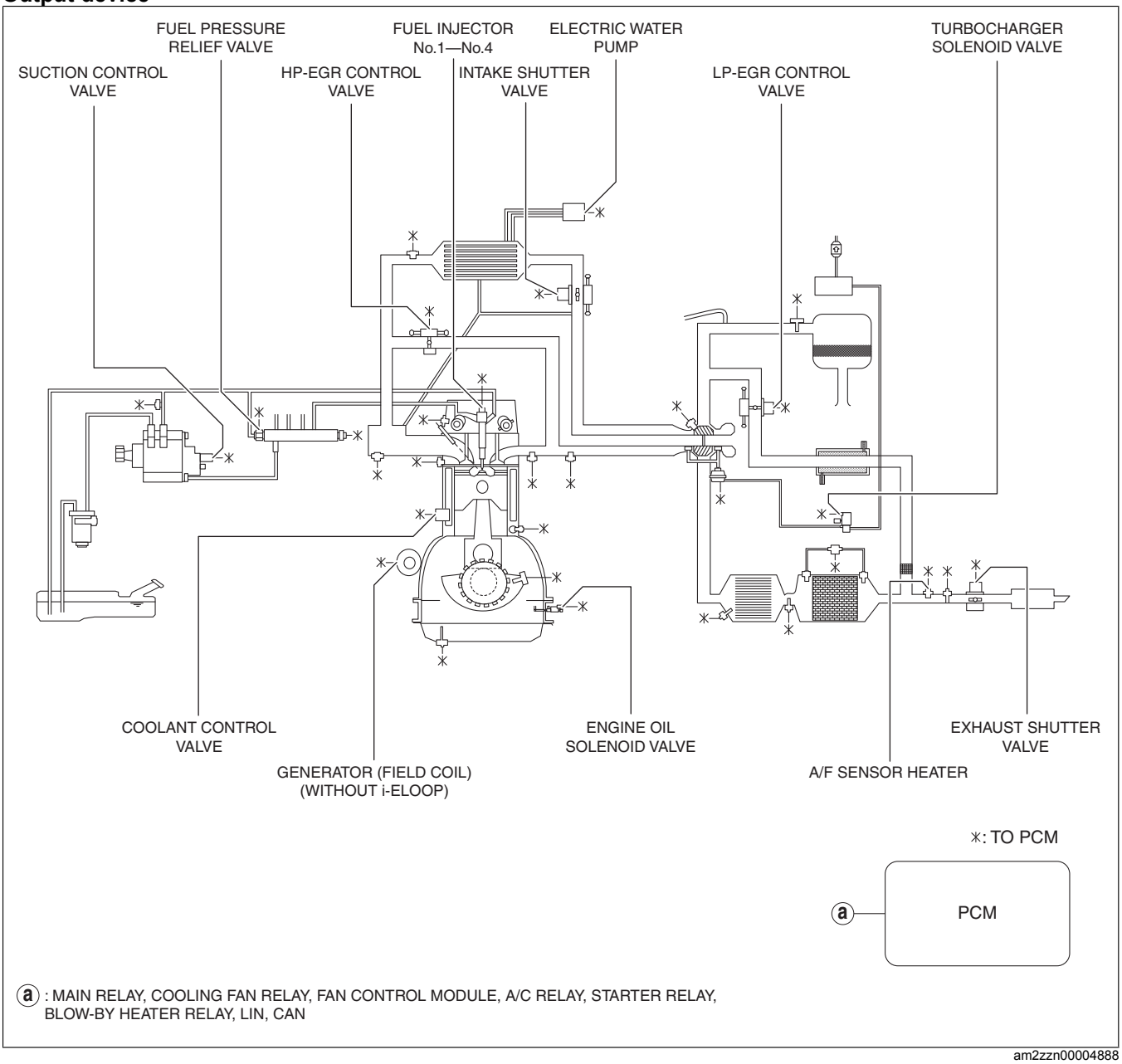


(a) : APP SENSOR No.1, No.2, POWER BRAKE UNIT VACUUM SENSOR, REFRIGERANT PRESSURE SENSOR, AMBIENT TEMPERATURE SENSOR, BATTERY, IG1 RELAY, CHECK CONNECTOR, BRAKE SWITCH (No.1 SIGNAL), (No.2 SIGNAL), CLUTCH STROKE SENSOR (MTX), STARTER INTERLOCK SWITCH (MTX), NEUTRAL SWITCH No.1 (MTX), NEUTRAL SWITCH No.2 (MTX), CPP SWITCH (MTX), BACK-UP LIGHT SWITCH (MTX), CAN, LIN

am2zzn00004887

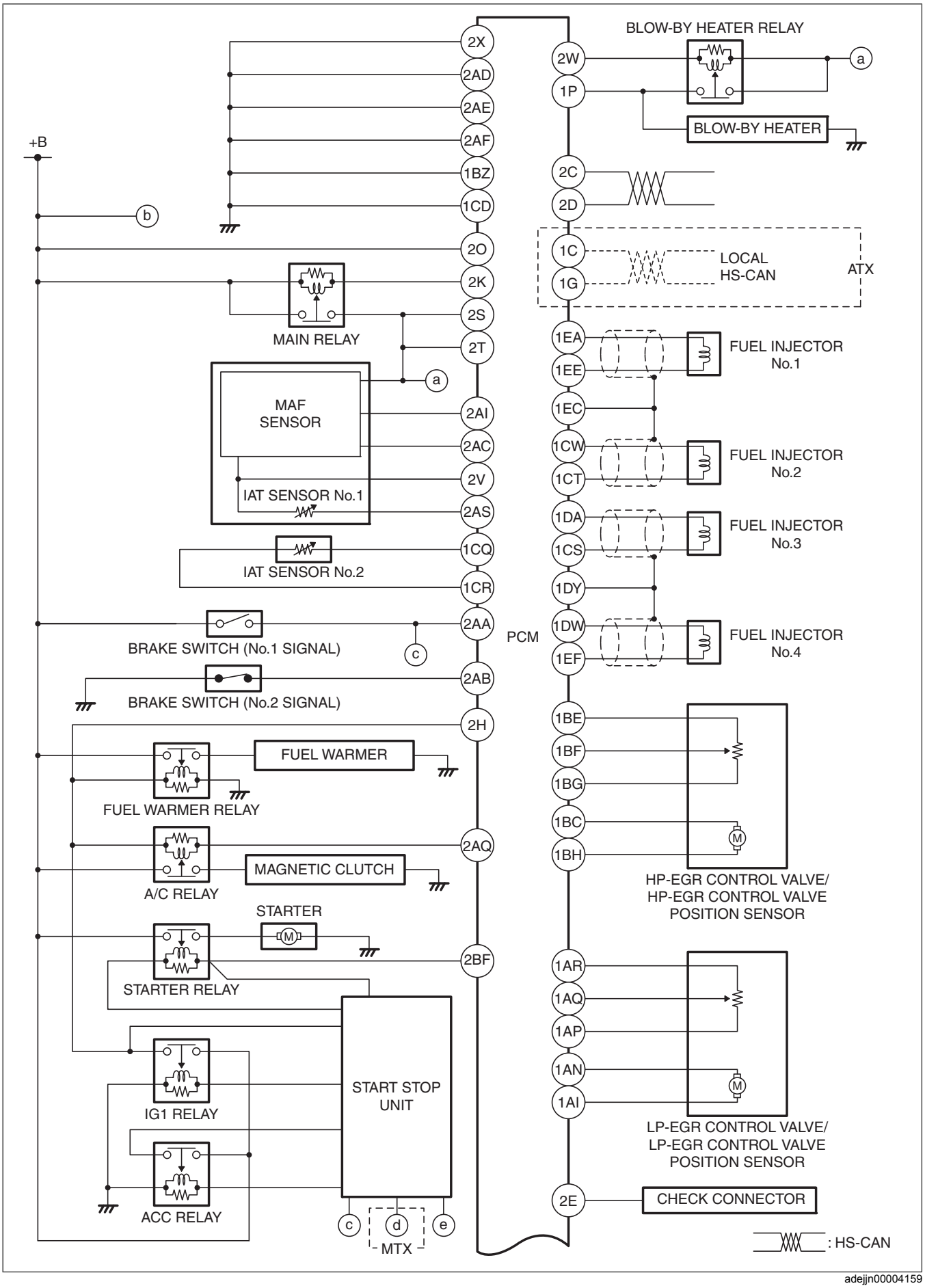
*1 : With engine oil level sensor

Output device

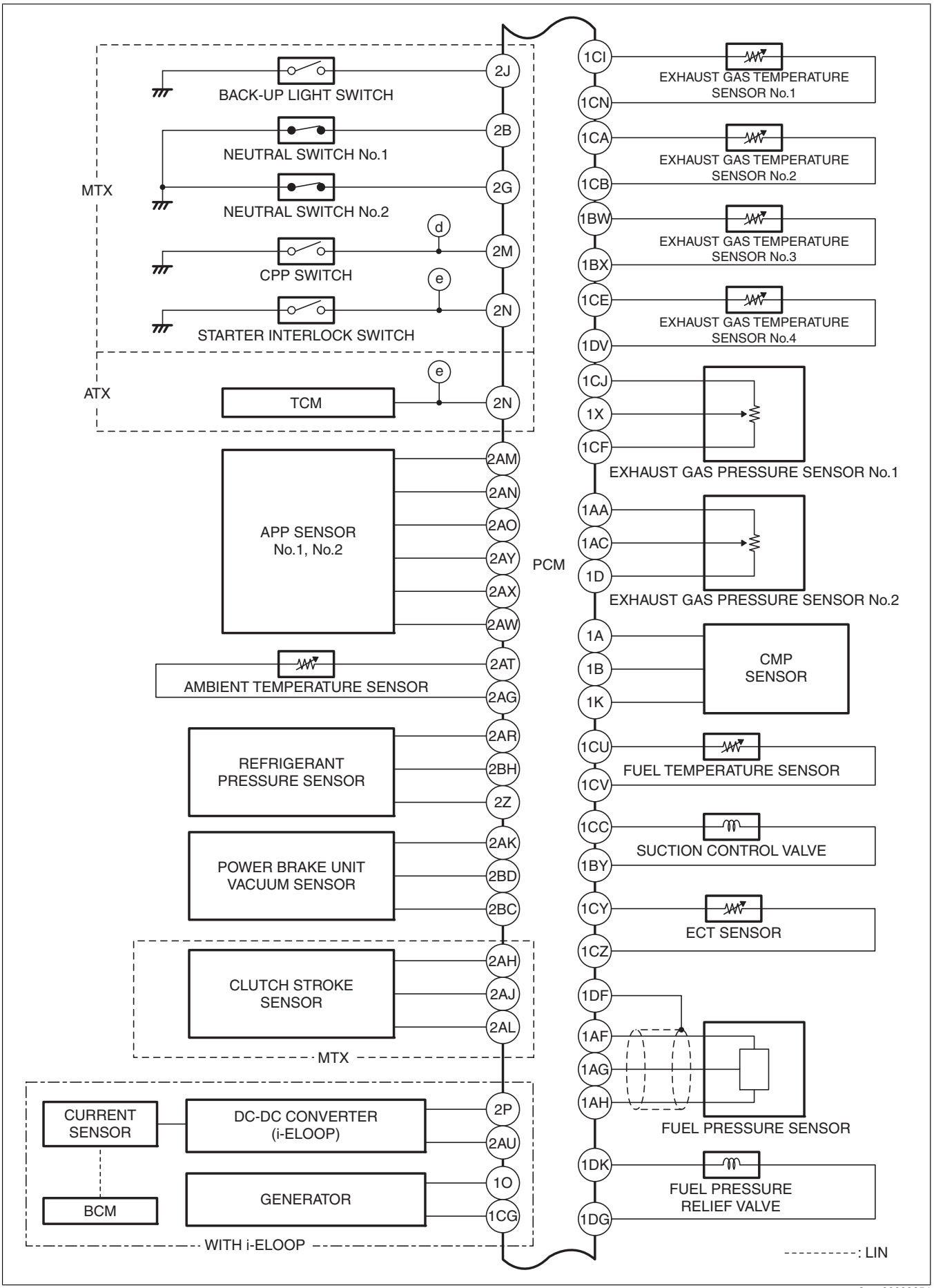


am2zzn00004888

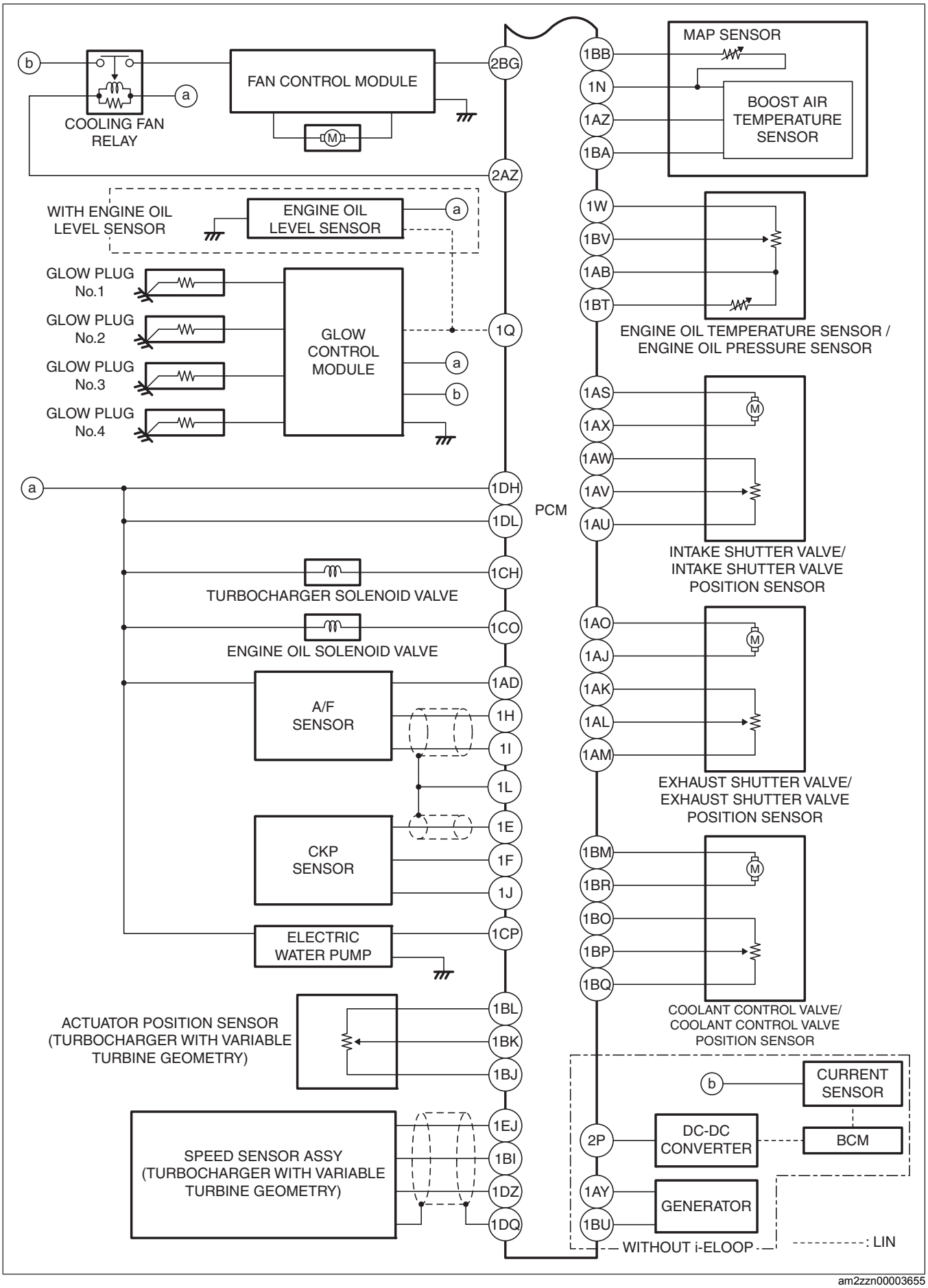
System Wiring Diagram



adejjn00004159

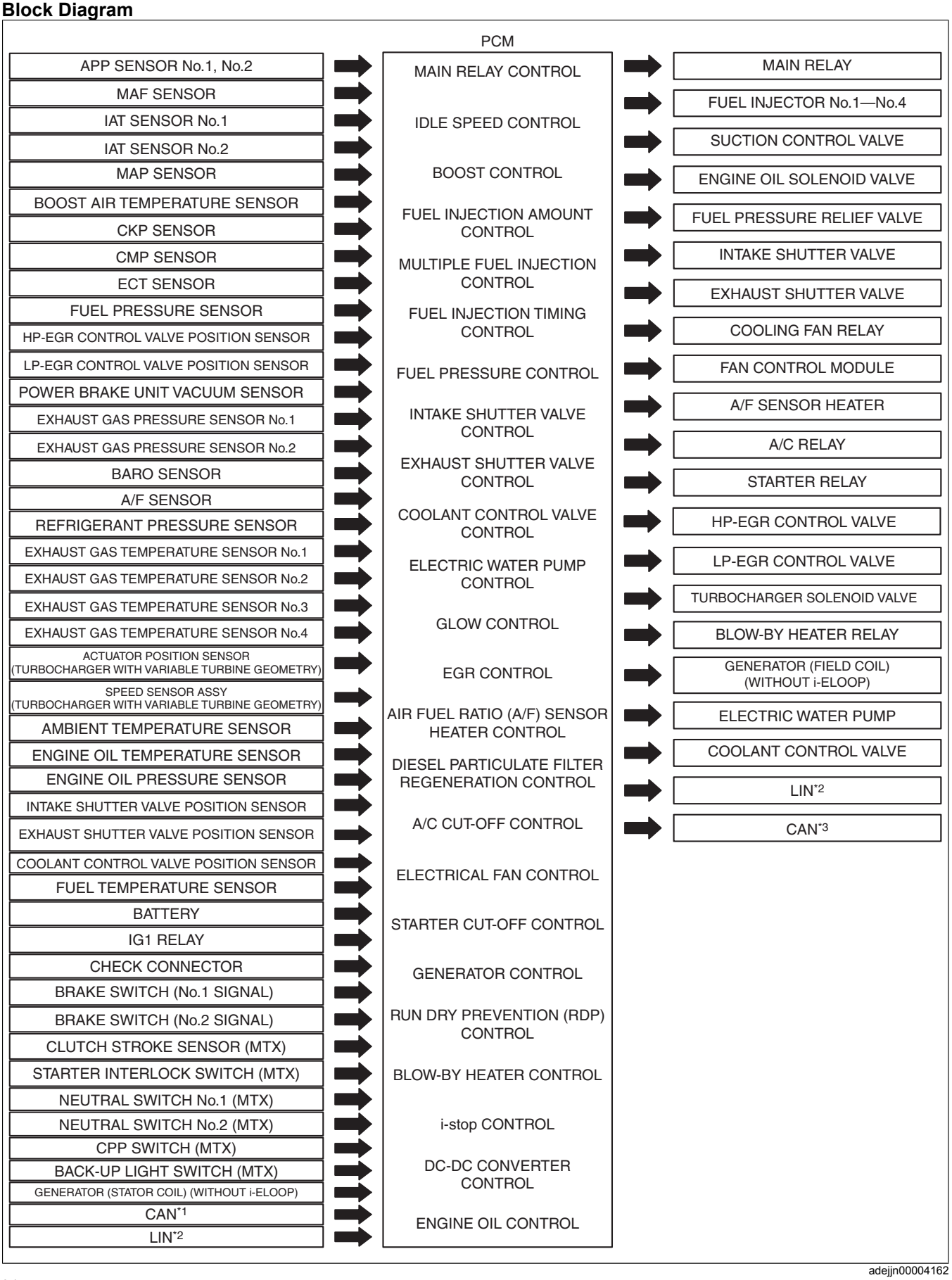


am2zzn00003654



am2zzn00003655

Block Diagram



adejjn00004162

*1 : DSC HU/CM, TCM (ATX), EPS control module, SAS control module, instrument cluster, BCM, start stop unit

*2 : Glow control module, engine oil level sensor

*3 : Instrument cluster (charging system warning light, master warning light, glow indicator light, diesel particulate filter indicator light, i-stop warning light (amber)/i-stop indicator light (green), i-stop indicator alarm/i-stop warning alarm)

Relation Chart

- Each control system and their related input and output parts are as follows.

×: Applicable

Item	MAIN RELAY CONTROL	IDLE SPEED CONTROL	BOOST CONTROL	FUEL INJECTION AMOUNT CONTROL	MULTIPLE FUEL INJECTION CONTROL	FUEL INJECTION TIMING CONTROL	FUEL PRESSURE CONTROL	INTAKE SHUTTER VALVE CONTROL	EXHAUST SHUTTER VALVE CONTROL	COOLANT CONTROL VALVE CONTROL	ELECTRIC WATER PUMP CONTROL	GLOW CONTROL
Input device												
Check connector					×							
IG1 relay	×		×	×	×	×		×	×	×	×	×
Brake switch (No.1 signal)												
Brake switch (No.2 signal)												
MAP sensor		×	×	×	×	×		×				×
Boost air temperature sensor											×	
MAF sensor		×	×	×	×	×	×	×	×			
IAT sensor No.1		×	×	×	×	×	×	×	×	×	×	×
IAT sensor No.2				×	×	×	×	×	×		×	
ECT sensor		×	×	×	×	×	×	×	×	×		×
Fuel temperature sensor				×	×	×	×					
APP sensor No. 1, No.2		×	×	×	×	×	×	×	×	×	×	×
Fuel pressure sensor		×		×	×	×	×					
Intake shutter valve position sensor		×						×				×
Exhaust shutter valve position sensor									×			
Coolant control valve position sensor										×		
HP-EGR control valve position sensor		×	×				×					×
LP-EGR control valve position sensor			×				×					×
Exhaust gas pressure sensor No.1			×	×	×	×	×					
Exhaust gas pressure sensor No.2				×	×	×	×					
BARO sensor		×	×	×	×	×	×	×	×			×
Battery			×							×	×	×

Item	MAIN RELAY CONTROL	IDLE SPEED CONTROL	BOOST CONTROL	FUEL INJECTION AMOUNT CONTROL	MULTIPLE FUEL INJECTION CONTROL	FUEL INJECTION TIMING CONTROL	FUEL PRESSURE CONTROL	INTAKE SHUTTER VALVE CONTROL	EXHAUST SHUTTER VALVE CONTROL	COOLANT CONTROL VALVE CONTROL	ELECTRIC WATER PUMP CONTROL	GLOW CONTROL
Refrigerant pressure sensor		×										
Exhaust gas temperature sensor No.1			×	×	×	×	×					
Exhaust gas temperature sensor No.2				×	×	×	×					
Exhaust gas temperature sensor No.3				×	×	×	×					
Exhaust gas temperature sensor No.4			×						×			
Engine oil temperature sensor										×		
Engine oil pressure sensor		×										
Actuator position sensor (turbocharger with variable turbine geometry)			×									
Speed sensor assy (turbocharger with variable turbine geometry)			×	×								
Ambient temperature sensor												
A/F sensor				×	×	×	×					
Power brake unit vacuum sensor												
CKP sensor	×	×	×	×	×	×	×	×	×	×	×	×
CMP sensor		×		×	×	×	×	×	×		×	
Back-up light switch (MTX)		×		×	×	×						
Neutral switch No.1 (MTX)		×	×	×	×	×						
CPP switch (MTX)		×		×	×	×		×	×			
Starter interlock switch (MTX)				×	×	×						
Neutral switch No.2 (MTX)				×	×	×						
Clutch stroke sensor (MTX)		×										
Generator (stator coil) (without i-ELOOP)				×	×	×	×					

Item	MAIN RELAY CONTROL	IDLE SPEED CONTROL	BOOST CONTROL	FUEL INJECTION AMOUNT CONTROL	MULTIPLE FUEL INJECTION CONTROL	FUEL INJECTION TIMING CONTROL	FUEL PRESSURE CONTROL	INTAKE SHUTTER VALVE CONTROL	EXHAUST SHUTTER VALVE CONTROL	COOLANT CONTROL VALVE CONTROL	ELECTRIC WATER PUMP CONTROL	GLOW CONTROL
CAN*1		×	×	×	×	×	×	×	×	×	×	×
LIN*2												×
Output device												
Intake shutter valve		×						×				
Exhaust shutter valve									×			
HP-EGR control valve		×										
LP-EGR control valve												
Fuel pressure relief valve							×					
Main relay	×											
Turbocharger solenoid valve			×									
A/F sensor heater												
Starter relay												
DC-DC converter (without i-ELOOP)												
Generator (Field coil) (without i-ELOOP)												
A/C relay												
Fan control module												
Cooling fan relay												
Engine oil solenoid valve		×										
Suction control valve		×					×					
Fuel injector No. 1—No.4		×		×	×	×						
Blow-by heater relay												
Electric water pump											×	
Coolant control valve										×		
LIN*2												×
CAN*3												×

*1 : DSC HU/CM, TCM (ATX), EPS control module, SAS control module, instrument cluster, BCM, start stop unit

*2 : Glow control module, engine oil level sensor

*3 : Instrument cluster (charging system warning light, master warning light, glow indicator light, diesel particulate filter indicator light, i-stop warning light (amber)/i-stop indicator light (green), i-stop indicator alarm/i-stop warning alarm)

×: Applicable

Item	EGR CONTROL	AIR FUEL RATIO (A/F) SENSOR HEATER CONTROL	DIESEL PARTICULATE FILTER REGENERATION CONTROL	A/C CUT-OFF CONTROL	ELECTRICAL FAN CONTROL	STARTER CUT-OFF CONTROL	GENERATOR CONTROL	RUN DRY PREVENTION (RDP) CONTROL	BLOW-BY HEATER CONTROL	i-stop CONTROL	DC-DC CONVERTER CONTROL	ENGINE OIL CONTROL
Input device												
Check connector			×									
IG1 relay				×	×	×				×	×	
Brake switch (No.1 signal)				×						×		
Brake switch (No.2 signal)										×		
MAP sensor	×		×									
Boost air temperature sensor			×									
MAF sensor	×		×									
IAT sensor No.1	×		×				×			×		
IAT sensor No.2	×	×	×									
ECT sensor	×	×	×	×	×	×				×		×
Fuel temperature sensor												
APP sensor No. 1, No.2	×		×	×						×		×
Fuel pressure sensor			×							×		
Intake shutter valve position sensor												
Exhaust shutter valve position sensor			×									
Coolant control valve position sensor												
HP-EGR control valve position sensor	×		×									
LP-EGR control valve position sensor	×		×									
Exhaust gas pressure sensor No.1	×		×									
Exhaust gas pressure sensor No.2			×									
BARO sensor	×		×							×		
Battery		×			×		×			×		×
Refrigerant pressure sensor				×	×							

Item	EGR CONTROL	AIR FUEL RATIO (A/F) SENSOR HEATER CONTROL	DIESEL PARTICULATE FILTER REGENERATION CONTROL	A/C CUT-OFF CONTROL	ELECTRICAL FAN CONTROL	STARTER CUT-OFF CONTROL	GENERATOR CONTROL	RUN DRY PREVENTION (RDP) CONTROL	BLOW-BY HEATER CONTROL	i-stop CONTROL	DC-DC CONVERTER CONTROL	ENGINE OIL CONTROL
Exhaust gas temperature sensor No.1	×		×									
Exhaust gas temperature sensor No.2			×									
Exhaust gas temperature sensor No.3			×		×							
Exhaust gas temperature sensor No.4	×	×	×									
Engine oil temperature sensor			×									×
Engine oil pressure sensor	×		×									
Actuator position sensor (turbocharger with variable turbine geometry)			×									
Speed sensor assy (turbocharger with variable turbine geometry)												
Ambient temperature sensor									×			
A/F sensor	×		×									
Power brake unit vacuum sensor										×		
CKP sensor	×	×	×	×		×	×		×	×		×
CMP sensor	×									×		
Back-up light switch (MTX)										×		
Neutral switch No.1 (MTX)			×	×		×				×		
CPP switch (MTX)	×		×							×		
Starter interlock switch (MTX)						×				×		
Neutral switch No.2 (MTX)										×		
Clutch stroke sensor (MTX)										×		

Item	EGR CONTROL	AIR FUEL RATIO (A/F) SENSOR HEATER CONTROL	DIESEL PARTICULATE FILTER REGENERATION CONTROL	A/C CUT-OFF CONTROL	ELECTRICAL FAN CONTROL	STARTER CUT-OFF CONTROL	GENERATOR CONTROL	RUN DRY PREVENTION (RDP) CONTROL	BLOW-BY HEATER CONTROL	i-stop CONTROL	DC-DC CONVERTER CONTROL	ENGINE OIL CONTROL
Generator (stator coil) (without i-ELOOP)							×					
CAN*1	×		×	×	×	×	×	×	×	×	×	×
LIN*2												
Output device												
Intake shutter valve			×							×		
Exhaust shutter valve										×		
HP-EGR control valve	×		×							×		
LP-EGR control valve	×		×							×		
Fuel pressure relief valve												
Main relay												
Turbocharger solenoid valve			×									
A/F sensor heater		×										
Starter relay						×				×		
DC-DC converter (without i-ELOOP)											×	
Generator (Field coil) (without i-ELOOP)							×			×		
A/C relay				×						×		
Fan control module					×							
Cooling fan relay					×							
Engine oil solenoid valve			×									×
Suction control valve												
Fuel injector No. 1—No.4			×					×		×		
Blow-by heater relay									×			
Electric water pump												
Coolant control valve												
LIN*2			×							×		
CAN*3			×				×		×	×		

*1 : DSC HU/CM, TCM (ATX), EPS control module, SAS control module, instrument cluster, BCM, start stop unit

*2 : Glow control module, engine oil level sensor

*3 : Instrument cluster (charging system warning light, master warning light, glow indicator light, diesel particulate filter indicator light, i-stop warning light (amber)/i-stop indicator light (green), i-stop indicator alarm/i-stop warning alarm)